

Modify Servo Motor for Continuous Movement



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Servo motors are widely used in precise control applications, but most of them can rotate up to 180 degrees only. Presented here is a process for modifying servo motors so that these can rotate up to 360 degrees in either direction (clockwise and anticlockwise). This would permit the motors' use for wider applications without using additional controllers.

The modification of a servo motor can be done using the following components and tools:

1. Two 2.2k, $\pm 1\%$, 0805 resistors
2. Soldering iron
3. Multimeter
4. Wire cutter
5. Precision screwdriver

De-assembling the servo motor

1. Remove any tags or stickers from the body of motor and open its front cover by removing the screws holding the motor together, as shown in Fig. 1.

2. Slowly open the front cover. There is circuit board, potentiometer and motor inside the casing, as shown

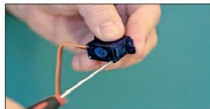


Fig. 1: Removing the front cover

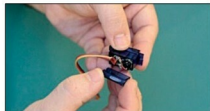


Fig. 2: Circuit inside the casing

in Fig. 2.

3. There are two sets of wires; one coming out of the motor and one coming out of the potentiometer. Cut the wires coming out of the potentiometer and do not damage the wires coming out of the motor, as shown in Fig. 3.

4. Open the top cover of motor to take out the potentiometer, as shown in Fig. 4.

5. Pay attention to the gear configuration (better take a photograph of it) as this will be needed during re-assembly, as shown in Fig. 5.

6. The knob shown in Fig. 6 helps limit the servo's movement.

7. Slowly pull the gear arrangement and potentiometer out of the casing. Try to minimise the movement between the gear arrangements. Cut the wires hanging from potentiometer, as shown in Fig. 7.

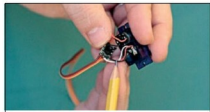


Fig. 3: Cutting the wires of potentiometer



Fig. 4: Opening top cover of the motor

Breaking the motion-hindering points

8. To break the motion-hindering points, the rim of the rivet needs to be removed. For doing so choose a drill bit which is a little bigger than the inner hole of the tiny rivet, as shown in Fig. 8.

9. While drilling, do not press hard and try not to get any grease on contact. Stop drilling as soon as doughnut-shaped rim of the rivet drills out, as shown in Fig. 9.



Fig. 5: Assembly of top part of the motor



Fig. 6: Knob of the motor



Fig. 7: Removing the gear arrangement of motor